

Tempest Spl Stake Pool

Security Assessment

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01 — Executive Summary

Overview

Tempest engaged OtterSec to assess the `spl-stake-pool` program. This assessment was conducted between January 2nd and January 11th, 2026. For more information on our auditing methodology, refer to Appendix B.

Key Findings

We produced 1 finding throughout this audit engagement.

In particular, we made suggestions regarding updating the codebase to remove unnecessary code and ensure proper documentation (OS-TSP-SUG-00).

Scope

The source code was delivered to us in a Git repository at https://github.com/Tempest-Finance/spl-stake-pool. This audit was performed against commit 8f705b3.

A brief description of the program is as follows:

Name	Description
spl-stake-pool	The audit focused on two instructions, DepositWsolWithSession and WithdrawWsolWithSession, that were added to the spl-stake-pool program to support wrapping and unwrapping WSOL when depositing or withdrawing SOL to or from the stake pool. The program targets the Fogo blockchain (SVM-compatible) and restricts these instructions to execution via Fogo sessions, analogous to account abstraction on Ethereum.

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02 — General Findings

Here, we present a discussion of general findings identified during our audit. While these findings do not pose an immediate security impact, they represent anti-patterns and may result in security issues in the future.

ID	Description
OS-TSP-SUG-00	Recommendation for updating the codebase to remove unnecessary code and ensure proper documentation.

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Code Maturity OS-TSP-SUG-00

Description

1. In `process_deposit_wsol_with_session`, subtracting `wsol_transiment_info.lamports`, while calculating the rent to refund is unnecessary since `wsol_transiment_info.lamports` will be zero after the account is closed. Thus, the subtraction operation may be removed.

>_ spl-stake-pool/program/src/processor.rs RUST

```rust
/// Processes [`DepositWsolWithSession`](enum.Instruction.html).
#[inline(never)]
fn process_deposit_wsol_with_session(
    program_id: &Pubkey,
    accounts: &[AccountInfo],
    deposit_lamports: u64,
    minimum_pool_tokens_out: Option<u64>,
) -> ProgramResult {
    [...]
    // Refund rent to payer
        let rent_lamports = rent
            .minimum_balance(spl_token::state::Account::LEN)
            .max(1)
            .saturating_sub(if wsol_transient_info.lamports() > 0 {
                wsol_transient_info.lamports()
            } else {
                0
            });
    [...]
}
```

2. The documentation for the account ordering of the `DepositWsolWithSession` instruction is incorrect, as index three is duplicated. Rectify the documentation.

Remediation

Implement the above-mentioned suggestions.

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A — Vulnerability Rating Scale

We rated our findings according to the following scale. Vulnerabilities have immediate security implications. Informational findings may be found in the General Findings.

CRITICAL

Vulnerabilities that immediately result in a loss of user funds with minimal preconditions.

Examples:

- Misconfigured authority or access control validation.
- Improperly designed economic incentives leading to loss of funds.

HIGH

Vulnerabilities that may result in a loss of user funds but are potentially difficult to exploit.

Examples:

- Loss of funds requiring specific victim interactions.
- Exploitation involving high capital requirement with respect to payout.

MEDIUM

Vulnerabilities that may result in denial of service scenarios or degraded usability.

Examples:

- Computational limit exhaustion through malicious input.
- Forced exceptions in the normal user flow.

LOW

Low probability vulnerabilities, which are still exploitable but require extenuating circumstances or undue risk.

Examples:

- Oracle manipulation with large capital requirements and multiple transactions.

INFO

Best practices to mitigate future security risks. These are classified as general findings.

Examples:

- Explicit assertion of critical internal invariants.
- Improved input validation.

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B — Procedure

As part of our standard auditing procedure, we split our analysis into two main sections: design and implementation.

When auditing the design of a program, we aim to ensure that the overall economic architecture is sound in the context of an on-chain program. In other words, there is no way to steal funds or deny service, ignoring any chain-specific quirks. This usually requires a deep understanding of the program's internal interactions, potential game theory implications, and general on-chain execution primitives.

One example of a design vulnerability would be an on-chain oracle that could be manipulated by flash loans or large deposits. Such a design would generally be unsound regardless of which chain the oracle is deployed on.

On the other hand, auditing the program's implementation requires a deep understanding of the chain's execution model. While this varies from chain to chain, some common implementation vulnerabilities include reentrancy, account ownership issues, arithmetic overflows, and rounding bugs.

As a general rule of thumb, implementation vulnerabilities tend to be more "checklist" style. In contrast, design vulnerabilities require a strong understanding of the underlying system and the various interactions: both with the user and cross-program.

As we approach any new target, we strive to comprehensively understand the program first. In our audits, we always approach targets with a team of auditors. This allows us to share thoughts and collaborate, picking up on details that others may have missed.

While sometimes the line between design and implementation can be blurry, we hope this gives some insight into our auditing procedure and thought process.

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